Analytical Chemistry 1 (125101)

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Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 gravimetric analysis (L12)

❖ Summary (L13)

2.4 Gran plot in Titrations

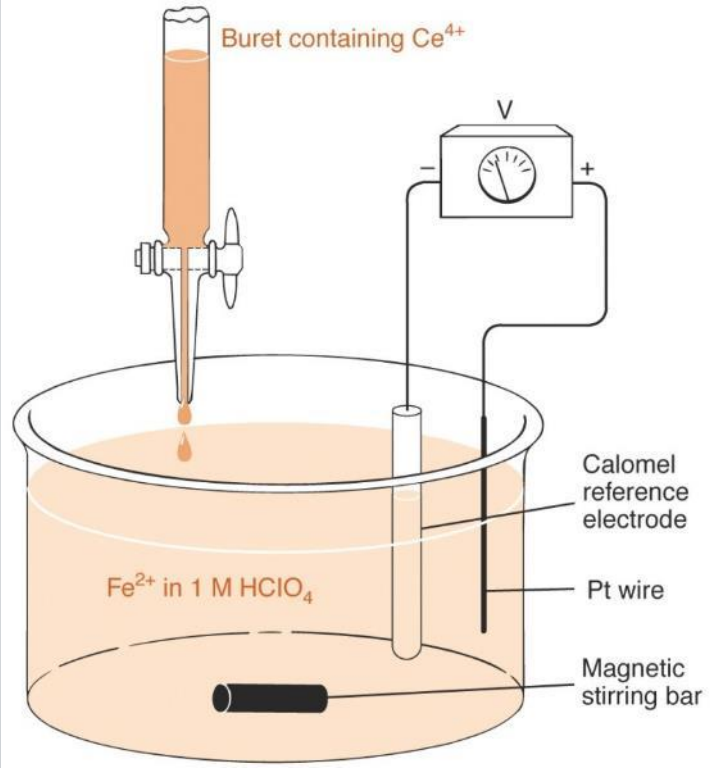
2.4 How to find the end point in titration

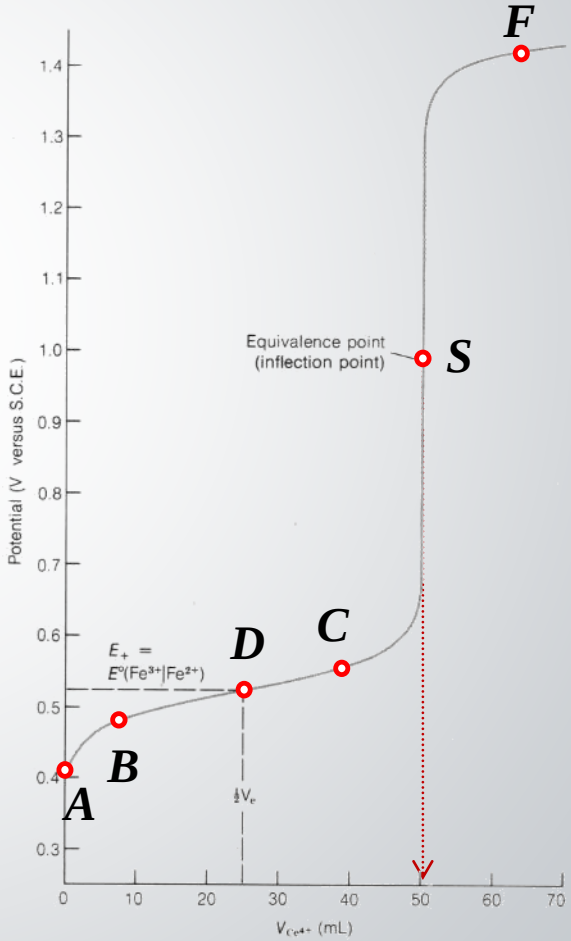
Overview

- Indicator method
- Mathematics methods
  - 2.4-1 Derivative method
  - 2.4-2 Gran plot method

Potentiometric on redox - titration

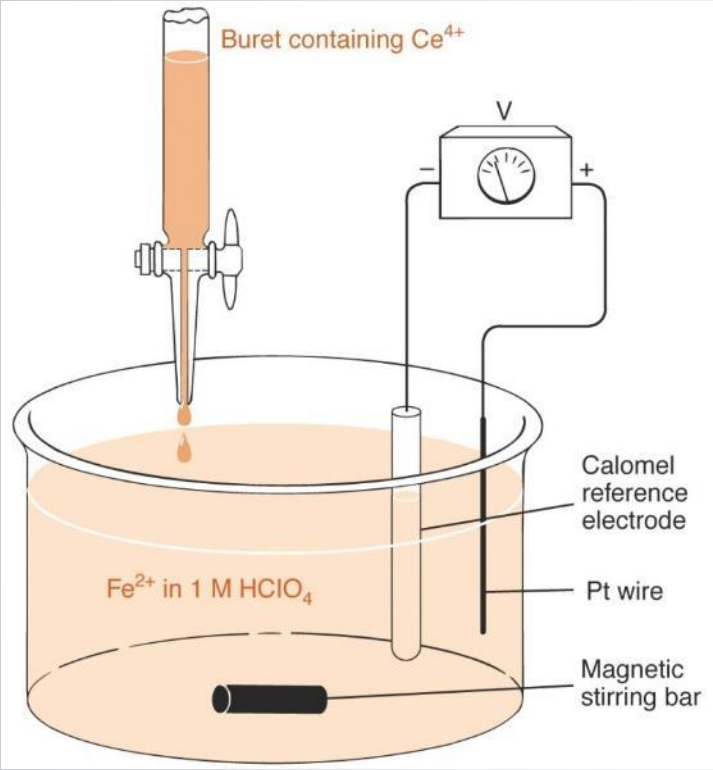
Application of working electrode for measuring potentials can be used for obtaining quantitative chemical information.





$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$

Potentiometric on redox - titration



25 mL of the Fe^{2+} solution was titrated with 0.1091 M Ce^{4+} with a solvent of 1 M HClO4. The cell was monitored potentiometrically with Pt and calomel electrodes.

What is the potential during the titration at 10% of the titration,

25%, 50%, 100%, 120% and 200% ?

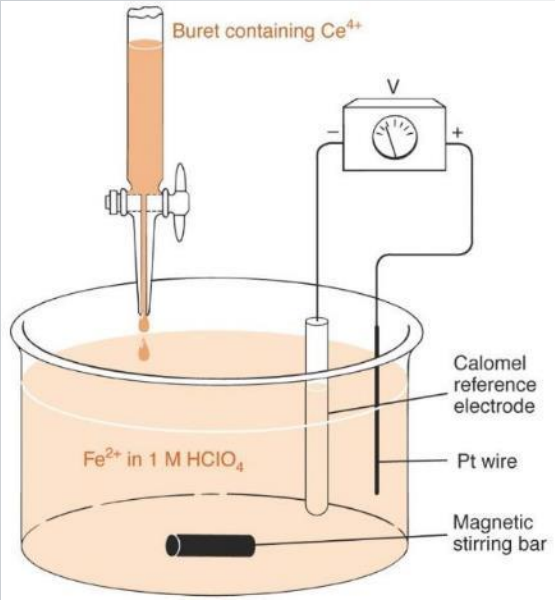
Titration reaction:

$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$ $K \approx 10^{16}$ in $HClO_4$

$$E^0(Fe^{3+}/Fe^{2+}) = 0.767\ V;\ E^0(Ce^{4+}/Ce^{3+}) = 1.70\ V;\ E^0(Ref.) = 0.241\ V$$

Potentiometric on redox - titration

At the Pt working electrode, two reactions will occur:



$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$

$$Fe^{3+} + e \rightleftarrows Fe^{2+} \quad E^0 = 0.767\ V$$

$$Ce^{4+} + e \rightleftarrows Ce^{3+} \quad E^0 = 1.70\ V$$

$$E_+ = E^0_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E_+ = E^0_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Ce^{3+}}{Ce^{4+}}$$

The final potential of the cell will be:

$$E(cell) = E_+ - E_-(calomel_electrode)$$

*Four regions for titration curve

1. Before equivalence point, 0< *v* <V_e

There is almost no Ce^{4+}, so it is better to work the potential use Fe element. We know how much Ce^{4+} was added which will give how much of the $[Fe^{3+}]$. For example, **10%** titration,

$$Ce^{4+} + Fe^{2+} \leftrightarrow Ce^{3+} + Fe^{3+}$$

$$0.9 \qquad\qquad 0.1$$

$$\frac{Fe^{2+}}{Fe^{3+}} = \frac{0.9}{0.1} = 9$$

$$E_{+} = E^{0}{}_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E(cell) = 0.767 - 0.05916\log(9) - 0.241 = 0.470V$$

Very tiny

Test yourself: How much of $[Ce^{4+}]$ in the solution? $[Ce^{4+}] = 1.95\times10^{-16}\%$

*Four regions for titration curve

1. Before equivalence point, 0< *v* <V_e

There is almost no Ce^{4+}, so it is better to work the potential use Fe element. For example, **25%** titration,

$$Ce^{4+} + Fe^{2+} \leftrightarrow Ce^{3+} + Fe^{3+}$$

$$0.75 \qquad\qquad 0.25$$

$$\frac{Fe^{2+}}{Fe^{3+}} = \frac{0.75}{0.25} = 3$$

$$E_{+} = E^{0}{}_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E(cell) = 0.767 - 0.05916\log(3) - 0.241 = 0.498V$$

*Four regions for titration curve

2. $v = \frac{1}{2}V_e$

There is almost no Ce^{4+}, so it is be better to work the potential use Fe element. For example, **50%** titration,

$$Ce^{4+} + Fe^{2+} \leftrightarrow Ce^{3+} + Fe^{3+}$$

$$0.5 \qquad\qquad 0.5$$

$$\frac{Fe^{2+}}{Fe^{3+}} = \frac{0.5}{0.5} = 1$$

$$E_{+} = E^{0}{}_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E(cell) = 0.767 - 0.241 = 0.526V$$

*Four regions for titration curve

3. At equivalence point, $v = V_e$ with 100% titration

Since the reaction is in equilibrium, a tiny amount of Fe^{2+} may be formed, which will be equal as that of Ce^{4+}, so we need to use both equations. Log *a* + log *b*= log (*ab),*

$$Ce^{4+} + Fe^{2+} \leftrightarrow \underset{1}{Ce^{3+}} + \underset{1}{Fe^{3+}}$$

$$E_{+} = E^{0}{}_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E_{+} = E^{0}{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Ce^{3+}}{Ce^{4+}}$$

$$2E_{+} = E^{0}{}_{Fe3+/Fe2+} + E^{0}{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{\cancel{Fe^{2+}}}{\cancel{Fe^{3+}}}\cdot\frac{\cancel{Ce^{3+}}}{\cancel{Ce^{4+}}}$$

*Four regions for titration curve

3. At equivalence point, $v = V_e$ with 100% titration

Since the reaction is in equilibrium, a tiny amount of Fe^{2+} may be formed, which will be equal as that of Ce^{4+}, so we need to use both equations. Log *a* + log *b*= log *ab*

$$2E_{+} = E^{0}_{Fe3+/Fe2+} + E^{0}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}\cdot\frac{Ce^{3+}}{Ce^{4+}}$$

$$2E_{+} = E^{0}_{Fe3+/Fe2+} + E^{0}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log(1)$$

$$E_{+} = \frac{(0.767 + 1.70)}{2} = 1.234V$$

$$E(cell) = 1.234 - 0.241 = 0.993\ V$$

So, working potential at equivalence point is the average of the two standard reduction potentials of analyte and titrant, which is independent of the concentrations and volumes of the reactants.

*Four regions for titration curve

4. After equivalence point, $v > V_e$

we don't know the $[Fe^{2+}]$, which has been "used up". There are excess of the Ce^{4+} and we know the quotient of $[Ce^{3+}]/[Ce^{4+}]$, e.g. **120%** titration,

$$Ce^{4+} + Fe^{2+} \leftrightarrow Ce^{3+} + Fe^{3+}$$

$$0.2 \qquad\qquad 1 \qquad 1$$

$$E(cell) = E^{0}{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Ce^{3+}}{Ce^{4+}} - 0.241$$

$$E(cell) = 1.70 - \frac{0.05916}{1}\log(5) - 0.241 = 1.418V$$

*Four regions for titration curve

4. After equivalence point, $v > V_e$

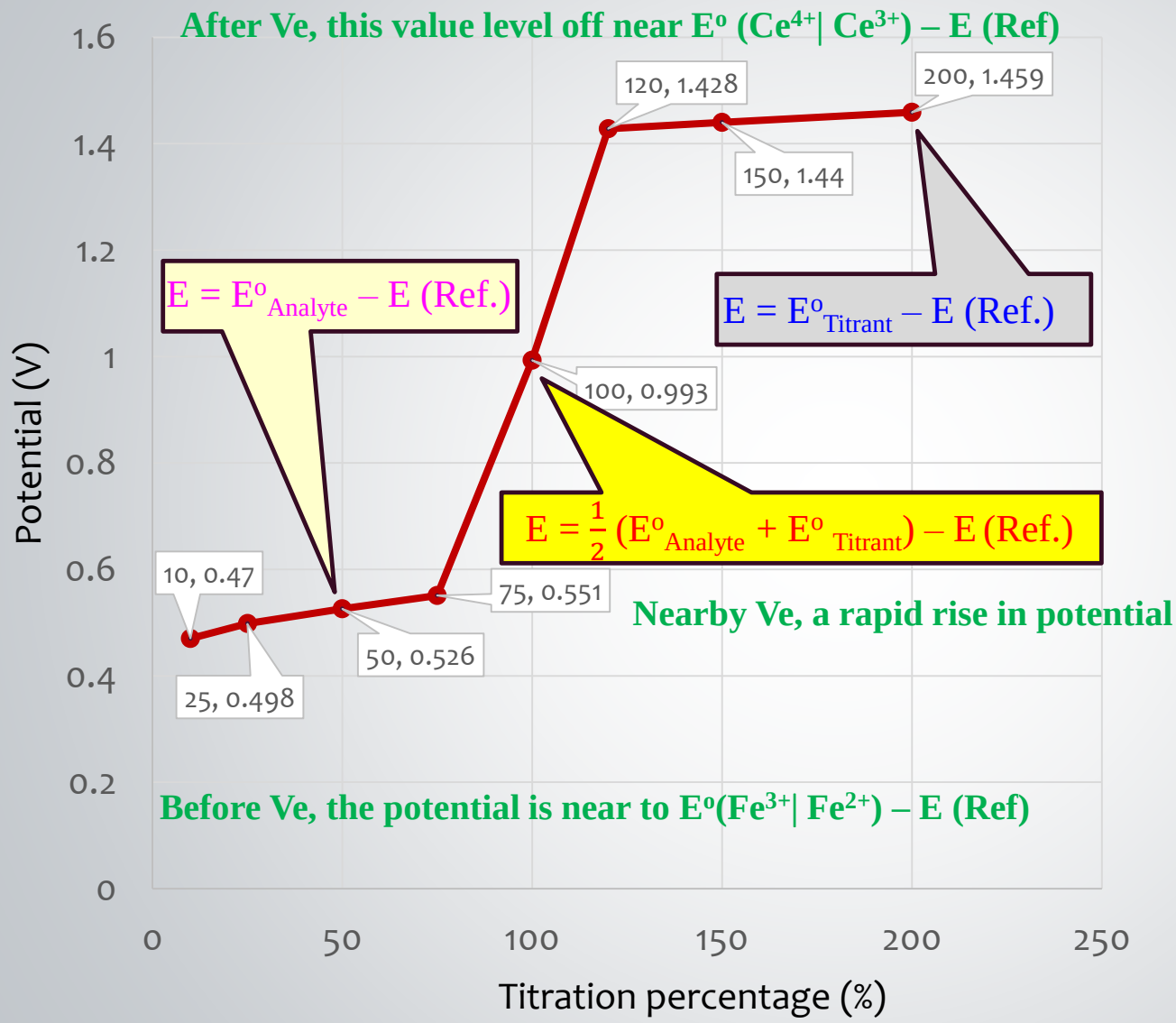
we don't know the $[Fe^{2+}]$, which has been "used up". There are excess of the Ce^{+} and we know the quotient of $[Ce^{3+}]/[Ce^{4+}]$, e.g. **200%** titration, the value is 1.

$$\underset{1}{Ce^{4+}} + Fe^{2+} \leftrightarrow \underset{1}{Ce^{3+}} + \underset{1}{Fe^{3+}}$$

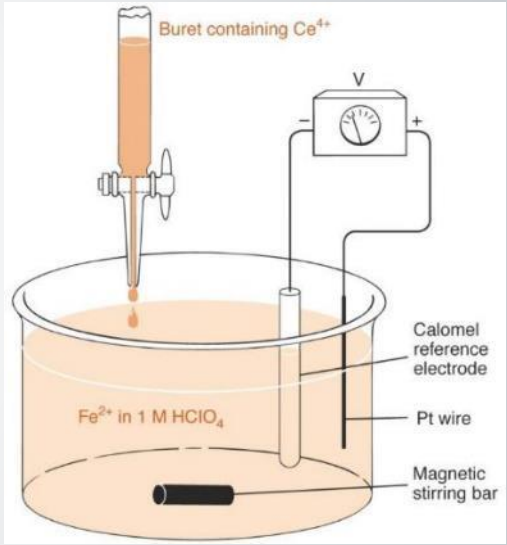
$$E(cell) = E^{0}{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Ce^{3+}}{Ce^{4+}} - 0.241$$

$$E(cell) = 1.70 - \frac{0.05916}{1}\log(1) - 0.241 = 1.459V$$

(b) Potentiometric on redox - titration



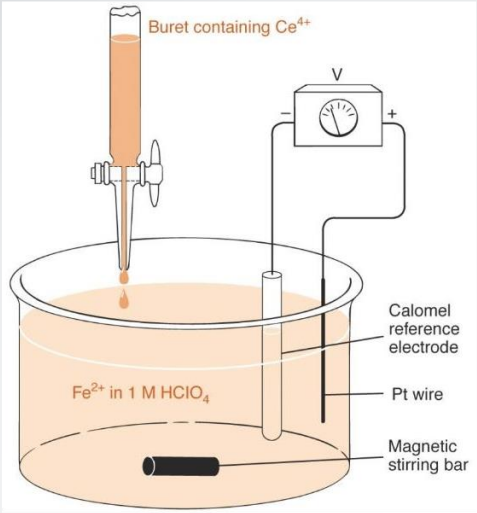
E° (Fe^{3+}/Fe^{2+}) = 0.767 V
E° (Ce^{4+}/Ce^{3+}) = 1.70 V
E (Ref.) = 0.241 V

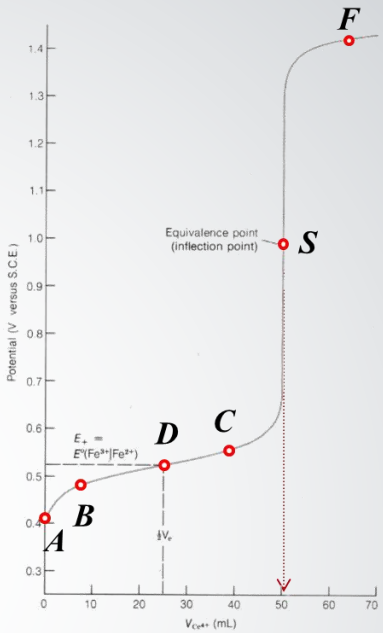


$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$

The larger ΔE between E°(analyte) and E°(titrant), the larger titration break, the error will be less, the titration accuracy would be better.

(b) Potentiometric on redox-titration



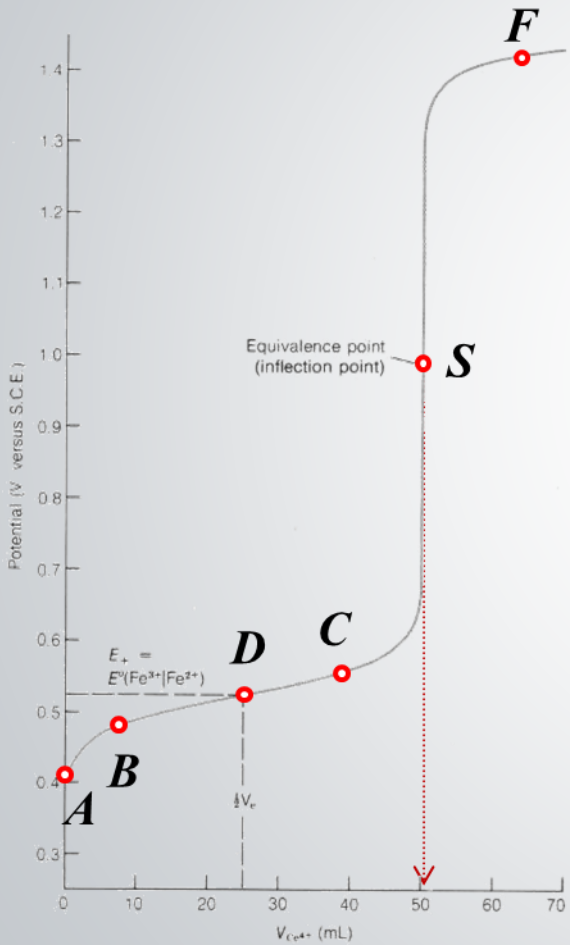


$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$

1) Why the titration curve goes up?
2) How about the E when titration is 0% (point A)?
3) Will the potentiometry influence the titration?
4) How to find the end point by Gran plot?

1) Why the titration curve goes up?

At the Pt working electrode, two reactions will occur:



$$Fe^{3+} + e \rightleftarrows Fe^{2+} \quad E^\circ = 0.767\ V$$

$$Ce^{4+} + e \rightleftarrows Ce^{3+} \quad E^\circ = 1.70\ V$$

Before equivalence point V < Ve (A to S)

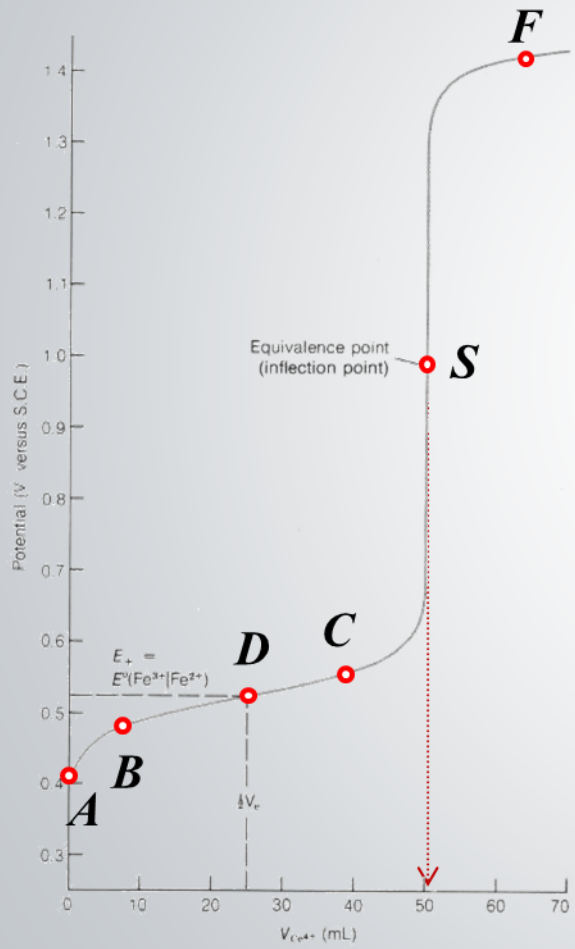
$$E_+ = E^0_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

The final potential of the cell is:

$$E(cell) = E_+ - E_-(calomel_electrode)$$

1) Why the titration curve goes up?

At the Pt working electrode, two reactions will occur:



$$Fe^{3+} + e \rightleftarrows Fe^{2+} \quad E^0 = 0.767\ V$$

$$Ce^{4+} + e \rightleftarrows Ce^{3+} \quad E^0 = 1.70\ V$$

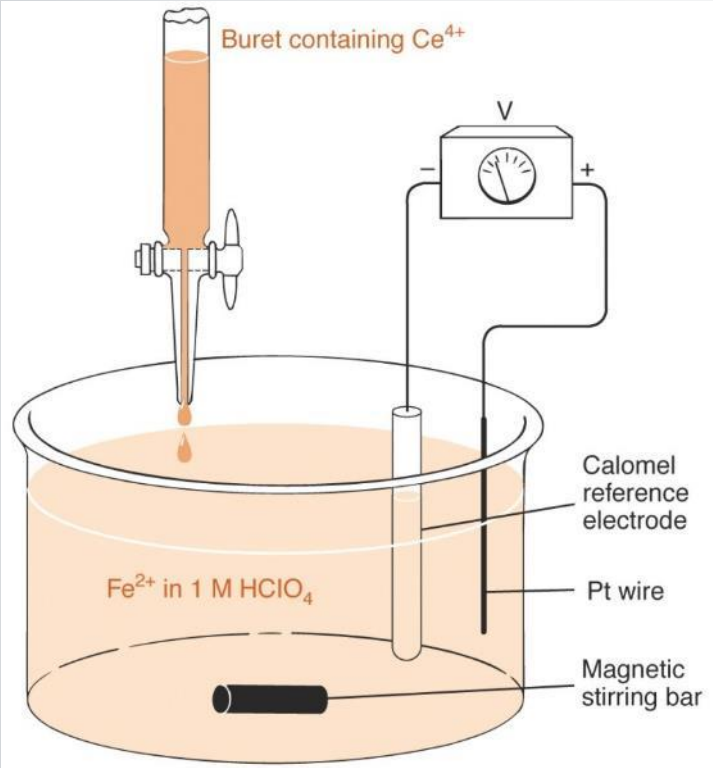
After equivalence point V > Ve (S to F)

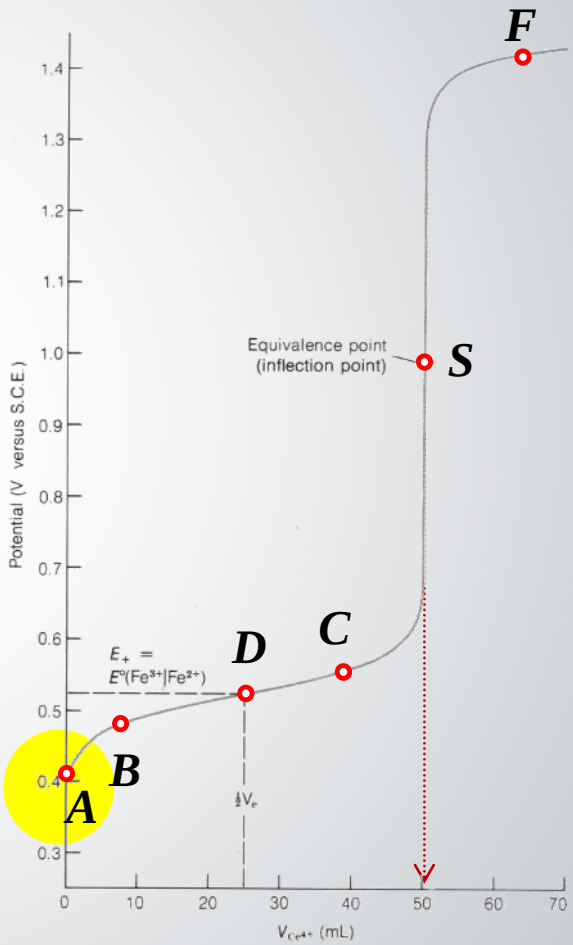
$$E_+ = E^0_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Ce^{3+}}{Ce^{4+}}$$

The final potential of the cell is:

$$E(cell) = E_+ - E_-(calomel_electrode)$$

2) How about the titration is 0% (point A)?





$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$

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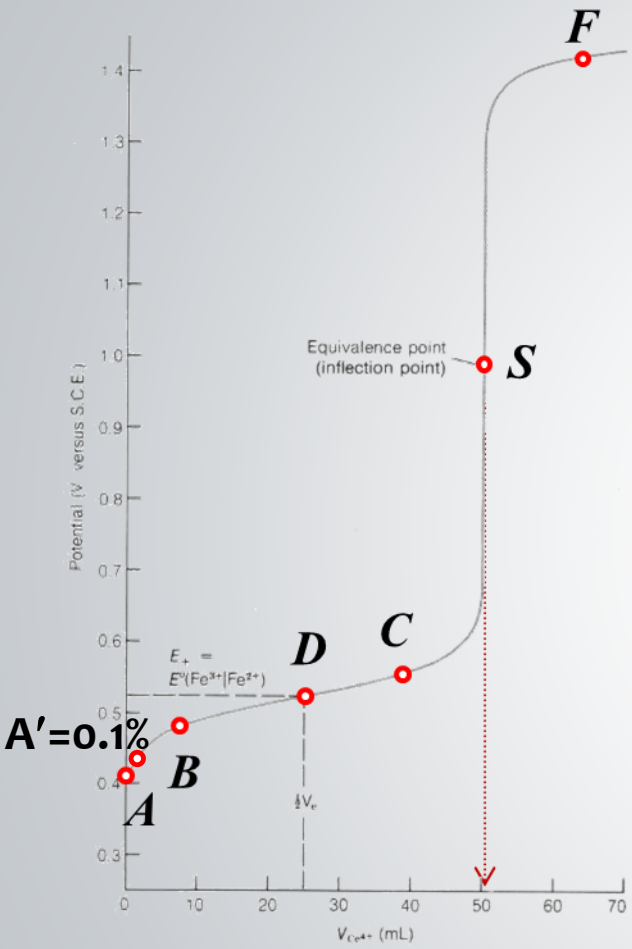


$$Fe^{3+} + e \rightleftarrows Fe^{2+} \quad E^0 = 0.767\ V$$

$$Ce^{4+} + e \rightleftarrows Ce^{3+} \quad E^0 = 1.70\ V$$

Before equivalence point V < Ve (point A)

$$E_+ = E^0_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

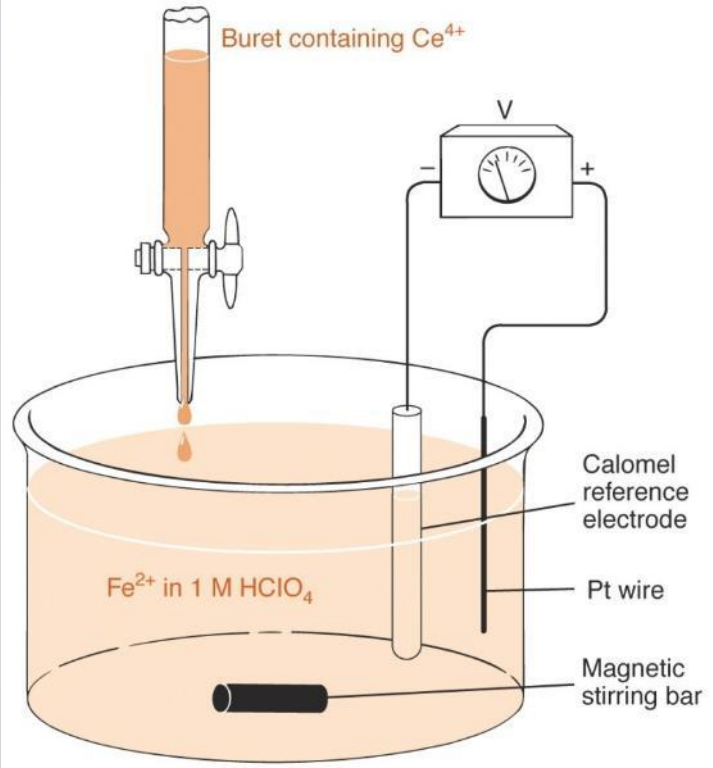
→ 100%

→ 0% ?

The voltage at V = 0 mL (point A) cannot be calculated, because we don't know how much $[Fe^{3+}]$. If $[Fe^{3+}]$ = 0 M, the voltage is -∞.

There must be some Fe^{3+}, either as an impurity or from oxidation of Fe^{2+} by atmospheric oxygen. The voltage could never be lower than that needed to reduce the solvent ($H_2O + e \rightarrow ½\ H_2 + OH^-$)

3) Will the potentiometry influence the titration?



At the Pt working electrode, two reactions will occur:

$$Fe^{3+} + e \rightleftarrows Fe^{2+} \quad E^0 = 0.767\ V$$

$$Ce^{4+} + e \rightleftarrows Ce^{3+} \quad E^0 = 1.70\ V$$

$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$

3) Will the potentiometry influence the titration?

Cell potential is measured under conditions of negligible current flow. The resistance of potentiometer is $10^{13}\ \Omega$. If a cell produces 1 V, the current through the circuit is

$$I = \frac{E}{R} = 10^{-13}\ \text{A}$$

The electron flow (from Faraday constant $= 9.649 \times 10^4$ C/mol) is

$$\frac{10^{-13}\,\text{C/s}}{9.649\times10^{4}\,\text{C/mol}} = 10^{-18}\ \text{mol e/s}$$

- Which produces negligible oxidation and reduction of reagents in the cell;
- To measures the voltage by potential meter will NOT affect the concentration of reactants.

3) Will the potentiometry influence the titration?

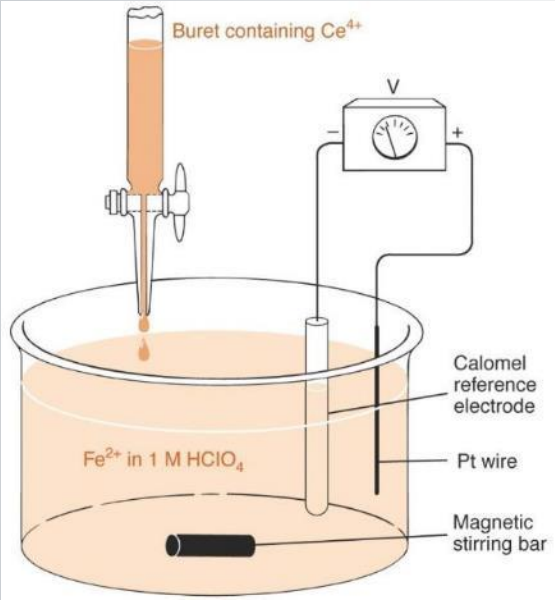
- If we replaces the potentiometer with a wire, there would be much more current and concentration would change until the cell reached equilibrium;
- At that point, nothing would drive the reaction, and the E would be 0 V. When the galvanic cell (battery) runs down to 0 V, the chemicals inside have reached equilibrium and the battery is "dead".

4) How to find the end point --- Gran plot

- Indicator and electrode are commonly used to find the end point in titration (e.g. redox, acid-base titration and precipitation titration);
- The end point is the maximum of the first derivative, $\frac{\Delta E}{\Delta V}$, or the zero crossing of the second derivative, $\frac{\Delta(\Delta E/\Delta V)}{\Delta V}$;
- The more accurate way to use titration data is to prepare a Gran plot. The Gran plot's method are derived from the data prior to the equivalence point.

Potentiometric on redox - titration

At the Pt working electrode, two reactions will occur:



$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$

$$Fe^{3+} + e \rightleftarrows Fe^{2+} \quad E^0 = 0.767\ V$$

$$Ce^{4+} + e \rightleftarrows Ce^{3+} \quad E^0 = 1.70\ V$$

Before equivalence point V < Ve:

$$E_+ = E^0_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

The final potential of the cell is:

$$E(cell) = E_+ - E_-(calomel_electrode)$$

4) How to find the end point --- Gran plot

Before equivalence point $V < V_e$:

$$E = E^o(Fe^{3+} \mid Fe^{2+}) - 0.05916\log\frac{[Fe2+]}{[Fe3+]} - E_{ref}$$

If the titrant volume is V and the reaction goes "to completion" with each addition of titrant, then

$$\frac{[Fe2+]}{[Fe3+]} = \frac{Ve - V}{V}$$

Substituting this expression into above equation and rearranging, eventually we obtain a linear equation of the form $y = kx + c$,

$$0.05916\log\frac{Ve - V}{V} = E^o - E_{ref} - E$$

$$\underbrace{V\times10^{-E/0.05916}}_{y} = \underbrace{V_e\times10^{(E_{ref} - E_0)/0.05916}}_{c} - \underbrace{V}_{x}\times\underbrace{10^{(E_{ref} - E_0)/0.05916}}_{k}$$

4) How to find the end point --- Gran plot

Before equivalence point $V < V_e$:

$$E = E^o(Fe^{3+} \mid Fe^{2+}) - 0.05916\log\frac{[Fe2+]}{[Fe3+]} - E_{ref}$$

$$\underbrace{V\times10^{-E/0.05916}}_{y} = \underbrace{V_e\times10^{(E_{ref} - E_0)/0.05916}}_{c} - \underbrace{V}_{x}\times\underbrace{10^{(E_{ref} - E_0)/0.05916}}_{k}$$

New method to find out the end point:

❖ A graph of $V\times10^{-E/0.05016}$ versus **V** should be a straight line with x-intercept is V_e.

$$E_{cell} = E^{\circ} - 0.05p \log \frac{[Fe^{2+}]}{[Fe^{3+}]} - E_{Ref}$$

$$0.05p \log \frac{V_e - V}{V} = -(E_{Ref} - E^{\circ}) - E_{cell}$$

$$\log \frac{V_e - V}{V} = -\frac{(E_{Ref} - E^{\circ})}{0.05p} - \frac{E_{cell}}{0.05p}$$

$$\frac{V_e - V}{V} = \frac{10^{-(E_{Ref} - E^{\circ})/0.05p} \cdot 10^{-E_{cell}/0.05p}}{\cancel{10^{E_{cell}/0.05p}}}$$

$$\frac{V_e - V}{V} = \frac{10^{-E_{cell}/0.05p}}{10^{(Ref - E^{\circ})/0.05p}}$$

$$V \cdot 10^{-E_{cell}/0.05p} = V_e \cdot 10^{(Ref - E^{\circ})/0.05p} - V \cdot 10^{(E_{Ref} - E^{\circ})/0.05p}$$

4) How to find the end point --- Gran plot

A titration of 50.0 mL of unknown Fe^{2+} with 0.100 M Ce^{4+} at 25°C, monitored with Pt and calomel electrodes, data before Ve are listed in the table,

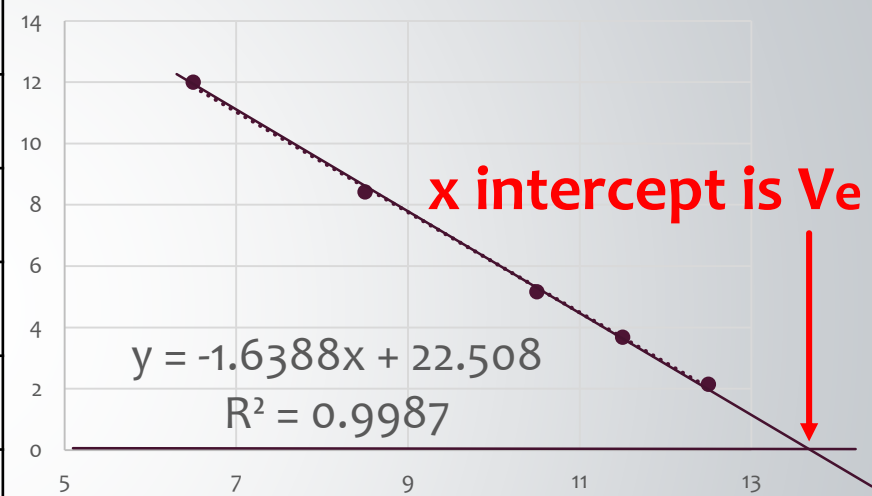
Titrant volume (mL)	E (volts)
6.50	0.635
8.50	0.651
10.50	0.669
11.50	0.680
12.50	0.696

Try to find the end point (Ve) for this redox titration by Gran plot and the unknown of [Fe^{2+}].

4) How to find the end point --- Gran plot

Computing the x-y in Gran plot of $V \times 10^{-E/0.05916}$ versus V,

Titrant volume (mL)	E (volts)	$10^{11} \times V \times 10^{-E/0.05916}$
6.50	0.635	12.003
8.50	0.651	8.4210
10.50	0.669	5.1626
11.50	0.680	3.6851
12.50	0.696	2.1488



Plot X-Y in Excel or calculate the k and c by least square to get the equation,

$$Y = -1.6388\,x + 22.508$$?

$Y = 0,\ X = V_e \Rightarrow V_e = 13.73\ \text{mL} \Rightarrow [Fe^{2+}] = \dfrac{0.100\ M \quad 13.73\ mL}{50.0\ mL} = 27.5\ \text{mM}$

4) How to find the end point --- Gran plot

❖ A problem with using derivatives to find the end point is that titration data are less accurate near the end point, because electrode response is sluggish nearby the equivalence point;

❖ A Gran plot uses data from before end point (typically from 0.8Ve to Ve) to locate the end point, which become more accurate.

Gran plot in acid-base titration

Try to use Gran plot to find end point in weak acid titrated with strong base. $[H+] = 10^{-pH}$ and $\frac{VaCa - VbCb}{Cb} = \frac{VaCa}{Cb} - Vb = Ve - Vb$ (due to VaCa = VeCb)

$$Ka = \frac{[H+][A-]}{[HA]}$$

$$[A^-] = \frac{VbCb}{Va+Vb}$$

$$[HA] = \frac{VaCa - VbCb}{Va+Vb}$$

$$\underbrace{Vb \times 10^{-pH}}_{y} = \underbrace{KaVe}_{c} - \underset{k}{Ka}\underset{x}{Vb}$$

- A graph of $Vb \times 10^{-pH}$ versus Vb is the Gran plot, in which a straight light with the x-intercept is Ve and the slope is Ka.
- The last 10-20% of volume prior to Ve is used for a gran plot.

Find the end point of Q4 in homework H7 by Gran plot.

Gran plot in acid-base titration

Find the end point in H7-4 by Gran plot.

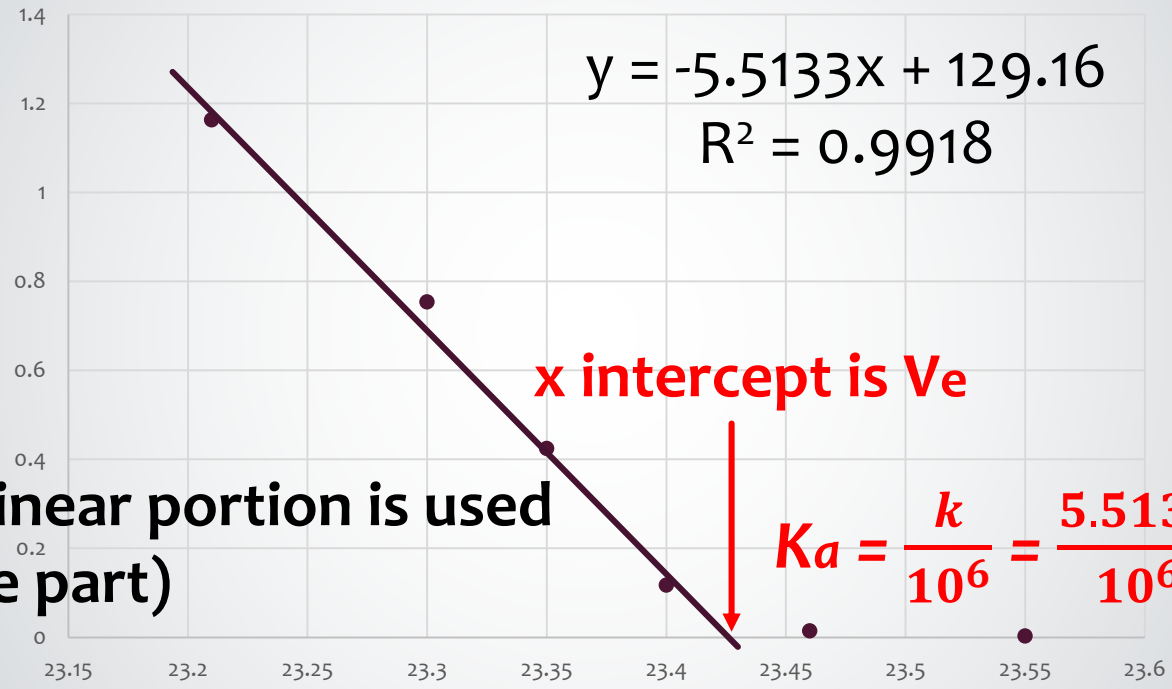
$$V_b \times 10^{-pH} = K_a V_e - K_a V_b$$

x (Vb)	$\bar{x} = \frac{x_1 + x_2}{2}$	y (pH)	$k = \frac{y_2 - y_1}{x_2 - x_1}$
23.21		7.30	
23.3	23.255	7.49	2.111
23.35	23.325	7.74	5
23.40	23.375	8.30	11.2
23.46	23.43	9.21	15.167
23.55	23.505	9.86	7.222

by 1st derivative

by Gran plot

Gran plot in acid-base titration for V_e and K_a



x intercept is Ve

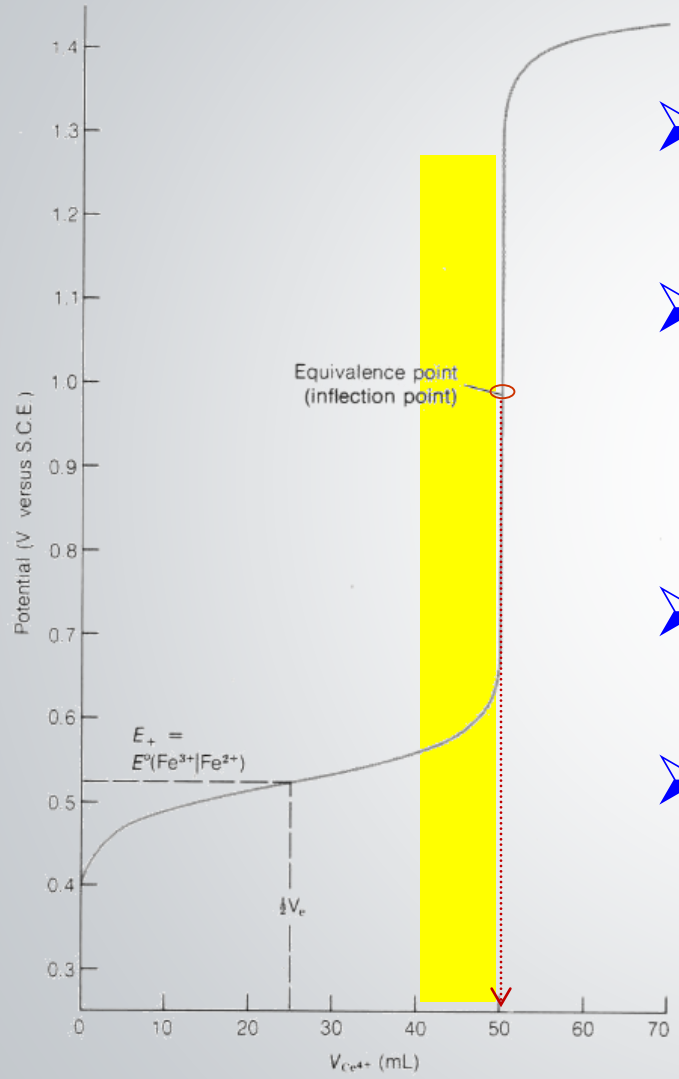
Only the linear portion is used (Before Ve part)

$$K_a = \frac{k}{10^6} = \frac{5.5133}{10^6} = 10^{-5.26}$$

The end point Ve = $\frac{129.16}{5.5133}$ mL = 23.43 mL by Gran plot;

Ve = 23.43 mL and Ka = $10^{-5.22}$ by first derivative method.

Summary of Gran plot in titration



- Gran plot used to find out Ve
- The concentration in the cell remain unchanged when we measure the potential during titration
- The potential at V=0 mL is unknown
- The titration curve raise during addition of titrant when E^o(Analyte) < E^o (titrant)

Gran plot data: 0.8Ve < V < Ve

Some important tips in Titration

- Understand the theory of acid-base, precipitation, and redox titration
- Calculation under a given volume of titrant could be done, e.g. $V < Ve$, $V = \frac{1}{2}Ve$, $V = Ve$, $V = 2Ve$ etc.
- Use different methods to find out end point, including how to choose a proper indicator, by first (second) derivative method, by Gran plot method.
- Know the effect of the concentration of reactants, Ka, Ksp, ΔE^{o}(Analyte | Titrant) on the titration error .